

Browser host (main thread, JavaScript) — the driver

Application / AI agent

ZetaDb · ZetaTxn · ZetaCursor
(JS API)

Web platform APIs — the “system interface”

OPFS

navigator.storage

WebCrypto

crypto.getRandomValues

performance

.now

query · execMut · begin · stream

persist / rehydrate

randomness

clock

wasm-bindgen binding layer (→target web) — the only boundary crossing

JS glue + injected host imports

WebAssembly sandbox — wasm32-unknown-unknown, single thread (no WASI, no Worker, no SharedArrayBuffer)

Engine JS-facing API surface

Zeta engine core (shared with the server build)

Parser → Planner → Optimizer

Vectorized executor +
streaming cursors

Log-centric async MVCC
versioned rows · timestamps · SI

Memory backend
(in-linear-memory tables + catalog)

Branches
= fork timestamps over the log

snapshot blob (bytes)

Snapshot codec
exportSnapshot /
openFromSnapshot

→ query path (JS API → engine)

- - -> capability binding — crosses the sandbox via wasm-bindgen and lands on a Web API, not a WASI syscall